Trading Strategies — Living Document

A running catalogue of every signal/strategy we've considered, built, tested, or shelved — with **status and evidence**, so we refine over time instead of re-litigating. Update the changelog at the bottom on every material change.

> **Golden rule:** nothing earns risk until it survives **point-in-time features + friction-aware, regime-split backtesting**. The broad-sentiment edge looked great and died exactly this way — see `backtest/engine.py`.

Status legend

	Meaning
✅ **Validated**	Survived friction-aware, regime-split backtest (still preliminary — see caveats)
🟡 **Marginal**	Real effect, but fragile net of costs / not sign-stable
❌ **Failed**	Tested and does not work (kept here so we don't retry it)
🧪 **Untested**	Built and wired, awaiting data to evaluate
💡 **Idea**	Backlog — not yet built
🔄 **In progress**	Data/build task underway

Progress at a glance

Last updated: 2026-06-03. Update this section whenever a status, metric, or next-step changes.

Strategy scoreboard

Strategy	Class	Status	Headline result	Next action
`retail_fear` (capitulation)	Macro overlay	❌ **FAILS the random-entry null**	Looked validated through excess/MT5/OOS/optimization — but vs **random entry** (same mechanics, random dates): **GOLD is 4th-percentile (WORSE than random; B&H +240% crushes it)**, AUDJPY only 76th (weak, n.s.). The apparent edge was **gold's drift**, not timing.	**do not deploy.** Re-examine only if a turn-based entry can beat random on a non-drifting instrument
`rrai_momentum_long`, `froth_high_long`	Macro overlay	❌ No edge	positive net but ~0 **excess vs buy-&-hold** (just bull drift)	dropped
`rrai_euphoria_short` + all macro shorts	Macro overlay	❌ Failed	−1.5 to −3pp excess (shorting the bull)	dropped
`dump_fade_DELETION_short`	Equity	❌ FAILS at full coverage (was a coverage-bias artifact)	After profiling **all 487 pump-authors (25%→100%)** the edge **flips negative**: gone≥0.20 → **−19.3%/10d (n=87)**, dominated by **squeeze tails** (IXHL −625%); PIT `young_frac` short is **−39% to −87%** (young pumps rip *hardest*). The earlier +14.3% (n=36 @ 25% cov) simply hadn't profiled the squeezers. Win rate still 64% but uncapped stock-short tails kill it.	**dead as a stock short**; only conceivable via **puts** (caps the −625% tail) — `distribution_short` is the better-behaved version (price-rolling-over filter dodges live squeezes)
`distribution_short` (concentrated pump + sentiment–price divergence)	Equity	✅ **The one validated edge — regime-CONDITIONAL**	**+11.78%/trade blended (t=3.57, PF 3.84, Sharpe 2.27); +20.96% in current regime (t=4.31, null 100th).** Generalizes micro→small→mid but **REVERSES on large/mega** (−1.9%, win 31% — Reddit can't move a $100B name); strongest in the **liquid $5-15 band (t=5.26)**. **MT5-validated on broker prices** (36/37 names tradeable, +11.02%, regime pattern replicates, 2025 +30%).	forward-OOS clock accruing; **trade liquid small/mid names via PUTS, tight-spread only**
`frothy_retail_regime` (regime filter / kill-switch)	Meta / regime	💡 **Idea — detector unbuilt**	distribution_short is regime-CONDITIONAL: it FIRES only in active retail froth, is STRONG when froth is **orderly** (2025-26 +21%), **DEAD when froth turns MANIC** (2021 re-squeezes, null 46th), and **dormant** in quiet regimes (2023 had ~12 concentrated pumps all year). The **signal count itself is a regime tell.**	build a robust mania detector — **froth-index FAILED** (2021≈2025), adaptive-performance only marginal; needs a small-cap squeeze-breadth / meme-vol index. Meanwhile: gate OFF in mania + always puts
`author_alpha` (predictive-user track records)	Equity / meta	✅ **Skill PERSISTS OOS — new edge**	`backtest/author_alpha.py`: rank authors on first-half calls → second-half **corr +0.27**; top-quintile future +0.58% vs **bottom −9.14%** (9.7pp OOS spread); 8.7% clear t>2 vs 2.5% chance. **Asymmetric — fade consistently-wrong authors** (stronger than following stars). High-n callers robust (henryzhangpku 7382/t=7.22); low-n leaders are luck.	build the **fade-the-bottom** backtest; weight pump signals by author quality; same micro-cap execution limits
`fade_organic_short`	Equity	🟠 Marginal	+0.83%/5d but P3 flips	retry via puts + VIX/vol gate

Strategy	Class	Status	Headline result	Next action
`pump_long` (ride)	Equity	❌ Failed	+3% but regime-flips, t=0.75	— drop
`dump_fade_short` (naive)	Equity	❌ Failed	−9.9% (squeezed)	superseded by deletion-gated
`organic_long`	Equity	❌ Failed	−4.2%, sign-stable loser	inverse → `fade_organic_short`

Build & data progress

Component	Status	Detail
Feature store (`feature_daily` / `aggregate_daily`)	✅ Done	~77k ticker-day rows, point-in-time
Friction-aware backtester (`backtest/engine.py`)	✅ Done	costs + regime split + no-lookahead
`/signals` feed (multi-strategy, json|mt5)	✅ Done	strategy-tagged; live, currently **flat**
Macro strategy R&D (`backtest/macro_research.py`)	✅ Done	only `retail_fear` beats buy-&-hold; momentum/froth no edge
MT5 framework (`SignalClient.mqh` + `RedditMacro_EA.mq5`)	✅ Done	strategy-selectable thin client; demo-only
Author profiling coverage	🔄 24% overall / **100% of pump-authors**	46,637 profiled, 6,153 gone; sweep now skips crypto-only + front-loads ticker-mentioners. **Targeted-profiled all 487 equity pump-authors.**
Deletion-gated short	❌ **Failed at full coverage**	The +14.3% was a coverage artifact (25% sample missed squeezers). At 100% pump-author coverage: gone≥0.20 **−19.3%/10d** (IXHL −625% tail); young_frac short −39%→−87%. Stock-short dead; puts-only if at all.
Short-interest / squeeze data (FINRA Reg SHO)	✅ Done	`scraper/finra.py` : 475k rows / 5,693 tickers; **z-score** screen (abs level is MM noise, median 0.49)
Independent fear-confirmation gate (Wikipedia)	🟡→🟢 Strengthened on 2016-26	**Extended series (525 caps, `backtest/fear_gate.py`): fear-only** excess SPY +0.57 (day t=1.64, **episode t=2.77**) / QQQ +0.80 (day t=1.91, **episode t=2.37**) — effect *grew* with the 2018 episodes & SPY/QQQ now **agree** ⇒ likely real (day-level still marginal). The LIVE **trend+fear still ≈0** (SPY +0.10/QQQ +0.11) → **trend filter should be dropped, use fear-only**. AUDJPY stays ungated (fear *hurts* it, +0.21→+0.09).
Live screens in `/signals` (`tradeable:false`)	✅ Done	`smallcap_pump_fade` · `fresh_pump_alert` · `coordination_flag` · `biotech_catalyst` · `attention_fade` · `calendar_overlay`
Idea backlog	💡 7 ideas	squeeze · options-flow · hype-cycle · sentiment-extremes · rotation · novelty · coordination
Equity/options paper client (Alpaca)	💡 Planned	not started

Methodology (non-negotiables)

1. **No lookahead** — features at date D use only data ≤ D; entries at the first close *after* D.
2. **Frictions** — round-trip spread (small-caps wide), short borrow, halts, capacity.
3. **Regime split** — net return per sub-period; sign-stability is the bar, not a big average.
4. **Shorts via PUTS**, not naked stock — penny shorts have squeeze/borrow/halt tails.
5. **Overlapping-window caveat** — daily signals with N-day forward returns inflate t-stats.

6. **Benchmark macro vs buy-and-hold** — in a bull market "long anything" looks like alpha; a macro long must beat *unconditional drift* (excess), not zero. (Cost us 5 false "survivors".)

Asset-class reality (what this data can/can't trade)

- **Small/micro-cap & meme equities** — primary signal (retail flow moves price).
- **Single-name options** — primary *expression* (convex longs, defined-risk shorts).
- **Risk-FX (AUD/NZD complex)** — the **primary macro expression**: retail capitulation → contrarian bounce, validated across the cross-section (AUDJPY/NZDJPY/AUDUSD/AUDCHF, t 2.7-3.2). The cleanest tradeable read of the risk-appetite factor RRAI measures.
- **Index / metals** — *macro overlay* (Gold robust; equities the weak expression, t≈1).
- **Haven/non-risk FX (USDJPY, CHFJPY, EURUSD) / "Reddit sentiment on SPY as direction"** — no edge; don't build it (they're the controls that confirm the factor).

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1. Macro overlay — Retail Risk-Appetite Index (RRAI)

Aggregate retail positioning as a **contrarian** factor at extremes. Built posts-only + ratios (coverage-stationary); 90-day trailing percentile. `analytics/features.py::aggregate_daily`.

> **Macro R&D verdict (`backtest/macro_research.py`, excess over buy-and-hold).** A bull market made *every* macro long look like a winner and *every* short a loser — until benchmarked against unconditional drift. Net of that, only the **contrarian retail-fear** family beat buy-and-hold. RRAI-momentum and froth had ~0 excess (pure drift); all shorts were −1.5 to −3pp (fighting the trend). **One real macro edge, three formulations of it.**

✅ `retail_fear` — buy risk when retail capitulates *(served live)*

- **Thesis:** retail max-bearish (RRAI pct low / bear-share extreme) + genuine fear (VIX high) → contrarian bounce. Captured 3 ways (capitulation, bear-extreme, pessimism-fade) — all positive-excess, mutually corroborating.
- **Spec (shipped):** long when `rrai_pct ≤ 0.15` **AND** `VIX ≥ 18`; hold ~10d; size by extremity; **fear-side only**.
- **Evidence (12mo, net, 10d): excess over buy-&-hold** SPY +0.17pp / QQQ +0.23pp / **AUDJPY +0.47pp** (win 88%); `bear_extreme` +0.24/+0.32/+0.36pp corroborates. **VIX gate ~4×'d it** (SPY +1.56%/10d in fear vs +0.42% calm). Modest **timing tilt**, not standalone alpha (buy-&-hold already won 66–75%).
- **Instrument scan (17 MT5 instruments, 10d excess vs buy-&-hold):** the signal **generalises across the risk-on basket** — Silver +2.22pp (lumpy, one big rally → size small), Gold +0.59pp (robust, +ve every regime), AUDJPY +0.47pp (steadiest, win 88%), SP500 +0.17pp. Oil (+1.82pp but one-episode) and crypto (BTC net-negative) **dropped**. Generalisation across assets is itself evidence it's a real risk-appetite signal.
- **★ Cross-sectional FX validation (2026-06-03, effective 2018-26 VIX-gated — the VIX≥18 gate has *no* signals in the calm 2016-17, so it starts Feb-2018/Volmageddon) — proves it's a RISK-ON-CURRENCY factor, not an AUDJPY fluke.** Ran capitulation-long across the yen-cross + risk-pair complex. The edge travels with the **risk-on currency (AUD/NZD)**, *independent of the funding leg* — and the **controls fail**, which is the real proof:

pair	character	excess	t(exc)
NZDJPY	risk-on JPY cross	+0.43%	**+3.18**
AUDJPY	risk-on JPY cross	+0.48%	**+2.93**
AUDCHF	risk vs CHF-haven	+0.41%	**+2.83**
AUDUSD	risk pair (no JPY)	+0.43%	**+2.72**
EURJPY	risk-ish	+0.25%	+2.70
NZDUSD	risk pair	+0.35%	+2.33
Nikkei ^N225	high-beta JP *equity*	+0.62%	+1.93
USDJPY	JPY cross, USD~haven	+0.12%	**+1.02** (flat)
CHFJPY	CONTROL haven/haven	+0.13%	**+1.37** (flat)
EURUSD	CONTROL non-risk	+0.14%	**+1.44** (flat)

Read: every AUD/NZD pair is significant (t 2.7–3.2) regardless of funding leg; the three pairs with *no* risk-on leg (USDJPY/CHFJPY/EURUSD) are the three weakest. **It's not a JPY thing** — JPY is just a clean funding leg; the Nikkei works because it's high-beta *risk*, not because it's Japanese. Equities stay the weak expression (SPY t=0.90, QQQ t=1.13). The clean cross-section (right pairs fire, controls don't) is strong evidence of a genuine factor → **trade the basket, not one pair.** - 🟡 **Independent fear-confirmation gate (Wikipedia, LIVE — but now DOUBTFUL):** the **equity** legs (US500/USTEC) additionally require an *independent* fear spike — `fear_z ≥ 0.5`, the z-score of Wikipedia fear-page attention (Stock_market_crash / Recession / Bear_market views) vs the trailing ~30d — **AND** a 200-DMA uptrend. Original rationale: SPY capitulation *alone* doesn't beat buy-and-hold (master backtest excess −0.05); an *ad-hoc* run reported **+0.36pp/10d when `fear_z` confirms vs ~−0.01 ungated**, and fear-attention is **1.32× higher on capitulation days** (`DATA_PLAN.md`). Wired live in `analytics/signals.py` (`_fear_z`, `FEAR_MIN`; `fear_z` in `/signals`). - 🔬 **MT5 Strategy-Tester A/B (US500, D1, 2020–2026, long-only, real spread/swap; `mql5/RedditMacro_US500_*.ini`, dates via `python main.py --export-signals`).** This tests the gate as an **absolute long-only rule** (NOT excess-vs-buy-&-hold):

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| variant | signals | trades | win% | net $ | PF | maxDD% |
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| baseline (all caps) | 335 | 100 | 57.0 | **+12** | 1.00 | 13.6 |
| `fear_z≥0.5` only | 95 | 45 | 53.3 | **-1045** | 0.68 | 13.0 |
| trend up (200-DMA) only | 223 | 71 | 56.3 | **+21** | 1.01 | 9.0 |
| trend **AND** fear (the live gate) | 54 | 27 | 55.6 | **-372** | 0.73 | 7.1 |

**Result: the `fear_z` gate does NOT help — it hurts.** Adding fear makes US500 *worse* both alone (-$1045)
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- ✅ **Committed EXCESS study (`backtest/fear_gate.py`, `python -m backtest.fear_gate`) — this settles it.** The frame MT5 can't measure: mean EXCESS over buy-&-hold (the +0.36pp's own metric). Same point-in-time date-sets as the MT5 run (shared `capitulation_sets()`). 10d, net of costs:

leg	variant	n	excess%	win%	t(exc)
SPY	baseline	332	−0.05	61	−0.21
SPY	**fear**	94	**+0.32**	68	+0.64
SPY	trend	220	−0.03	60	−0.16
SPY	trend+fear *(live)*	53	+0.09	64	+0.19
QQQ	baseline	332	+0.06	62	+0.20
QQQ	**fear**	94	**+0.41**	65	+0.71
QQQ	trend+fear *(live)*	53	**−0.20**	60	−0.33
AUDJPY *(ungated)*	baseline	333	**+0.26**	68	**+2.04**

Verdict (three honest parts): (1) **The +0.36pp REPRODUCES as a point estimate** — fear-only SPY +0.32 / QQQ +0.41 excess — so it was *not* fabricated; the fear gate does lift equity-capitulation excess. (2) **But it is NOT statistically significant** (excess t≈0.6, n≈94) — a thin, unproven edge. (3) **The LIVE `trend+fear` combo is the wrong way to apply it**: the trend filter *removes the buy-fear-in-a-downtrend bounces that carry the excess*, diluting SPY (+0.32→+0.09) and flipping QQQ negative (+0.41→**−0.20**). - **Reconciliation (why MT5 said "loss" and this says "small win"):** MT5 sums *absolute* P&L and holds one position at a time, so a few COVID-cluster trades dominate the dollar total (−$1045); the excess frame means-averages independent trades, where fear is mildly +ve. Both are correct for their question — for *signal research* the excess frame is the right one, so the earlier "fear-gate is a drag" was a frame artifact. - **The genuine tension:** fear → better *timing/excess* but worse *drawdown* (COVID knives); trend → better *drawdown* but worse *excess*. The live `trend AND fear` gets the worst of both for excess. - **Recommendation:** keep the gate 🟡 (positive but insignificant). If we ever optimise the equity legs for excess, use **fear-only, not trend+fear**; AUDJPY (the one significant edge, +0.26 t=2.04) stays **ungated** — confirmed correct. (User chose to leave the live gate unchanged for now; this is logged for when we revisit.) - 🔢 **Power / effective sample (day vs episode — the divergence IS the verdict):** the 94 fear-days are *not* 94 independent obs — overlapping 10d windows + episode clusters. Treated as **days**, fear-only is t=0.64 (SPY)/0.71 (QQQ); collapsed to **independent episodes** (~40, one vote each) it's **t=2.28 (SPY)/1.85 (QQQ)**. The gap is driven almost entirely by the **COVID-2020 cluster** (~10+ losing fear-days → one episode vote), so the effect is *positive across most episodes but fragile* — and SPY (2.28) vs QQQ (1.85) disagreeing on near-identical instruments over the same episodes says we're at the **noise floor**, not a stable edge. **Data to settle it:** day-level needs ~**9× more** obs for p<0.05 (~19× for 80% power) — impractical (~40+ yrs at ~15-20 fear-days/yr); the real lever is **independent fear episodes**, and the cheapest source is *backward* (Wikipedia fear data already reaches 2015) — **extending the RRAI/capitulation series to 2016-2017 adds the 2018 Volmageddon + Q4-2018 episodes** (price coverage starts 2016-05). - ✅ **DONE — RRAI extended to 2016-06 (525 caps, +190).** Re-ran the study on the longer series, and the fear-gate **strengthened** (effect *grew* out-of-sample — the hallmark of a real edge):

leg	fear excess (94d → 141d)	day t (→)	episode t (→)
SPY	+0.32 → **+0.57**	0.64 → 1.64	2.28 → **2.77**
QQQ	+0.41 → **+0.80**	0.71 → 1.91	1.85 → **2.37**

Both equity legs are now episode-level significant (t≈2.4-2.8) AND agree — the earlier SPY≠QQQ noise-floor red flag is gone. Day-level is still only marginal (1.6-1.9), so not fully nailed, but this is a real upgrade from "fragile/unproven." The trend filter conclusion **hardens**: `trend+fear` (live) is still ≈0 excess (SPY +0.10/QQQ +0.11) ⇒ **drop `trend`, use fear-only** on the equity legs. AUDJPY confirmed best ungated (baseline +0.21 t=2.00; fear-gating *hurts* → +0.09). - **Expression (GOLD-LED basket, finalised 2026-06-03 after MT5+OOS): Gold (XAUUSD) primary + AUDJPY small secondary.** The excess t-stats liked the whole AUD/NZD complex, but realistic execution + OOS kept only these two: **AUDUSD dropped** (failed OOS, PF 0.47), **NZDJPY dropped** (no OOS), US500/USTEC/Silver dropped earlier (no alpha). `euphoria_short` parked. Served via `/signals` (tag `retail_fear`); traded by `mql5/RedditMacro_EA.mq5` with **per-instrument params** — Gold StopATR 3 / hold 11 / risk 1.0%; AUDJPY StopATR 2 / hold 24 / risk 0.5% (its OOS-optimal long hold, half size as a secondary). - **Caveats:** single ~12mo bull regime; **46 cap-days = only ~13 distinct fear episodes** (clustered → effective n small); overlapping windows; edge small in absolute terms. - **Next:** leverage-language enrichment (calls/puts, margin/YOLO); longer history (data-limited); vol-target sizing; combine capitulation+bear-extreme into one composite trigger.

❌ Macro candidates that failed (kept so we don't retry)

- `rrai_euphoria_short`, `bull_extreme_short`, `froth_high_short`, all momentum-shorts: **−1.5 to −3pp excess** — shorting a bull market.
- `rrai_up_momentum_long`, `froth_high_long`: **~0 excess** — just drift, no signal.

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2. Equity cross-sectional (per ticker × day)

From `analytics/features.py::feature_daily` (velocity, accel, breadth, concentration/HHI, sentiment trajectory, author young/gone/suspended mix, novelty).

❌ `pump_long` — ride concentrated promotion

- **Thesis:** concentrated mention spike (few authors, high volume) pops short-term.
- **Result:** net +3%/5d but **sign-flips** (P1 +15.8 → P3 −4.6), win 36%, t=0.75. The crude "+6.45%" was a no-lookahead/no-cost/single-regime artifact. **Not robust.**

❌ `dump_fade_short` — short a decelerating pump (naive)

- **Result:** net **−9.9%** (P1 −29%). Shorting pumps on deceleration gets squeezed/run over. Wrong trigger; brutal without defined risk.

❌ `organic_long` — buy broad attention spikes

- **Result:** sign-stable **−4.2%** net (t=−4.25, n=321). Buying broad/news-driven attention tops is a *consistent loser* → the inverse is the candidate (below).

🟡 `fade_organic_short` — fade broad attention spikes

- **Thesis:** broad attention spikes mean-revert down (inverse of the robust `organic_long` loss).
- **Result:** net +0.83%, but **flips in P3** (win 46%) — the reversion is real but shorting costs + recent weakening eat it. **Marginal.** Revisit via **puts** (cheaper than borrow) and with a VIX/vol gate.

❌ `dump_fade_DELETION_short` — short a pump whose promoters vanished *(THE thesis — DIED at full coverage)*

> ⚠️ **REVERSAL (2026-06-03).** This was "forensically validated" at +14.3%/10d (n=30-36) — but that ran at only **~25% pump-author coverage**. We then **targeted-profiled all 487 equity pump-authors (→100%)** and re-tested: the edge **flips to −19.3%/10d (n=87)**, and the PIT `young_frac` short is −39% to −87%. Cause: **squeeze tails** — a single ticker (IXHL) squeezed **−625%/−404%/−377%** in July-2025; the partial sample simply hadn't profiled its authors, so those events were invisible. Lesson logged: **small-n + partial-coverage results are dangerous**; shorting live manipulated micro-caps has *uncapped* left-tail risk. Win rate is still ~64% (most pumps fade), so the thesis isn't *wrong* — but it's only expressible via **puts** (defined risk caps the −625%). The price-rolling-over filter is what makes `distribution_short` survive where this dies. - **Thesis (original):** concentrated pump + promoters now **deleted/suspended** → manufactured → fade the collapse. The original A2 idea, and the *reason* we track account lifecycle. - **Result (2026-06-02, 20% author coverage, h=10, market-neutral, 75bps + 8bps/day borrow):** | gate | n | net short | win | t | regimes | |---|---|---|---|---|---| | `gone_frac≥0.10` | 33 | **+10.0%** | 81.8% | 1.51 | −1.2 / +19.9 / +11.4 | | `gone_frac≥0.20` | 30 | **+14.3%** | 83.3% | **2.57** | +11.4 / +27.7 / +3.9 | | **control:** pumps w/ promoters *still here* (`gone=0`) | 202 | **−10.8%** | 36.1% | −2.22 | −30.7 / −0.6 / −1.1 | - The gate **flips the sign of the trade** (~+25pp spread). Un-gated concentrated pumps are *un-shortable* (they continue / squeeze); only the ones whose promoters vanish collapse. The deletion is doing the work, not pump-selection. Positive across all 3 regimes at gone≥0.20. - ⚠️ **Not yet point-in-time tradeable.** `gone_frac` uses *hindsight*: Reddit exposes no deletion timestamp, so "gone" means "gone as of our latest sweep" — unknowable at the pump date. The PIT-available leading proxy is `young_frac` (account age at post time), but it's currently too sparse to conclude (n=3–7, and the few events lean the wrong way). - **Path to tradeable:** (1) **finish the author sweep** (20%→40%+) to power the `young_frac` PIT test; (2) failing that, a **live deletion-monitor** rule — `author_status_revalidate` already detects deletions forward; short on deletion-*confirmation* within K days of a fresh pump, capturing the back half of the collapse. Express via **puts** (defined risk vs squeeze tails; many are HTB/no-borrow).

🟡 `distribution_short` — concentrated pump + sentiment-price divergence *(NEW, point-in-time tradeable)*

- **Thesis:** the "bagholder distribution" moment — a *manufactured* pump (concentrated chatter) where **price has started rolling over but the crowd is still euphoric**. Insiders distribute into the retail bid; sentiment lags price. Every input (`per_author` , `mentions` , `sentiment` , trailing price return) is known at signal time → **unlike the deletion gate, this is actually tradeable live.** (`backtest/microstructure.py`)
- **Rule:** `per_author≥2 AND mentions≥10 AND sentiment≥0.4 AND 5d-trailing-return<0` → SHORT. h=10, market-neutral, 75bps + 8bps/day borrow.
- **Result (2026-06-02): +7.2%/10d net, t=2.70, win 52%, n=60** across **39 distinct tickers**.
- **Date-clustered test** (1 obs/date, kills intra-day correlation): **+7.6%, t=2.52, 47 independent dates** — *not* a risk-off-day artifact (max 3 events on any date, spread over 12 months).
- **Dose-response** in sentiment (s≥0.3/0.4/0.5 → +3.9/+7.2/+11.1%) and **monotone in horizon** (h=5/10/20 → +5.6/+7.2/+9.5%, t rising to 2.92) — the hallmarks of a real effect, not a fit.
- **Survives 2× costs** (150bps spread + 16bps/day borrow → still +4.9%/10d).
- **Control:** same concentrated pumps *without* the divergence (price up) → **−4.1%** — the divergence is doing the work. Positive in all 3 sub-periods.
- **Why it works where the others don't:** the gate is *precise* — it only fires inside the manufactured subset. The broad high-attention universe **drifts flat-to-up** (gross ≈ +0.04%), so generic pump-shorting just pays costs (see negatives below); the edge requires isolating the distributing pumps, which both this and the deletion gate do.
- **Caveats / next:** n still modest (47 indep. dates); single ~12-mo window (sub-periods, not true bull/bear regimes); small-cap-heavy so no multi-regime price history; loose `per_author≥2` lets some ETFs/large-caps (QQQ/IWM) leak in — tighten to ≥3 and/or split small-cap vs index. **Collect forward OOS** (the clock starts now); express via **puts**. Not yet promoted to the live `/signals` feed.

❌ Negatives from the same study (recorded so we don't re-run them)

- **Broad-universe microstructure timing does NOT survive costs.** Recruitment-exhaustion, mention-deceleration, and sentiment-price divergence applied to the *whole* high-attention set (mentions_z≥1) are all **net-negative** — the universe drifts flat-to-up, so shorting it generically just bleeds the ~2.3% friction. Precision (concentration filter) is mandatory.

- **Recruitment-exhaustion alone = null.** New-author inflow rolling over is directionally supportive *within* concentrated pumps (+1.8% net, but t=0.33) and adds to the stack, but is not a standalone edge at current coverage. The new-author feature is built and kept.
- **Up-leg longs both fail.** (a) "Ride while recruiting + accelerating" (concentrated) netted +1.6% but win 27%, regimes [+21/−7/−9] — one big winner, otherwise negative. (b) "Attention-leads-price coil" (broad: recruitment surging while price still flat) was **net −1.4 to −1.9%** across horizons, sign-stable loser. Neither is a tradeable long; dropped.

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3. Idea backlog (💡 not yet built)

- **Squeeze setup (long):** Reddit squeeze chatter (Shortsqueeze/SqueezePlays) **+ short pressure** → squeezable. *Partially built:* `scraper/finra.py` + `squeeze_screen()` flag tickers with an anomalous **z-score jump** in daily short-ratio vs their own baseline (HTZ/GME currently surface). *Still needs* true SI%/float/days-to-cover (FINRA **bi-monthly short-interest** report) + borrow-fee for the level; the daily file's absolute level is market-maker noise. *Not yet backtested.*
- **Options-flow signal:** parse contract-level chatter (unusual_whales/options) → retail-gamma/positioning proxy. *Needs contract parsing.*
- **Hype-cycle lifecycle:** model attention as nascent→acceleration→peak→fade; long the *organic* acceleration, fade the *concentrated* peak. Breadth-vs-concentration is the discriminator.
- **Conditioned sentiment extremes (single-name):** linear sentiment is dead, but *extreme concentrated euphoria at a price high* (short) / *capitulation after a sharp drop* (bounce) may be non-linear edges.
- **Cross-sectional attention rotation:** rank meme universe by abnormal velocity z-score; long top-decile / short bottom-decile, beta-hedged with SPY; baskets from co-mention clusters.
- **Novelty / first-mention alpha:** the "discovery" moment (silent → sudden organic burst).
- **Coordination graph (short):** rigorous version of the deletion short — cluster density + temporal bursts + duplicate text + new-account% + gone%.

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4. Data gaps blocking strategies

Gap	Unblocks
Intraday prices (have EOD only)	timing pump/dump entries/exits (they move intraday)
Short interest / float / borrow fee	squeeze *level* (FINRA daily short-*volume* now in via `scraper/finra.py`, but it's MM noise at the level; need bi-monthly SI%/float + borrow)
Options chains / IV	options expression; options-flow signal
Longer history (data-limited to ~12mo)	true multi-regime validation of the RRAI overlay
Author coverage (20% → target ~40%+)	a *point-in-time tradeable* deletion short — thesis already forensically validated; PIT `young_frac` proxy needs more coverage

5. Execution stack

- `/signals` **feed** (`analytics/signals.py` , `GET /signals` json|mt5) — live, serves the capitulation overlay. ✅
- **MT5 EA** (`mql5/RedditRRAI_EA.mq5`) — thin client for the index/FX overlay; demo-only. ✅
- **Alpaca / IBKR paper client** — for the equity/options legs. 💡 planned.

★ MULTI-REGIME VALIDATION (2020-2026, incl. COVID crash + 2022 bear)

The deep-history backfill let us test across regimes — the result that actually moves confidence: - **AUDJPY capitulation-long is REGIME-ROBUST** — positive in EVERY year 2020-2026, **including the 2022 bear** (+0.7%, win 60%), win rates 60-93%. Graduates from preliminary → **validated**. This is the core tradeable edge. - **Equity (SPY) capitulation-long FAILS in a sustained bear** (2022: -0.7%/10d, win 43%) — falling-knife problem → **must gate with the 200-DMA trend filter** (don't buy fear in a downtrend). Confirmed needed. - **Euphoria-short only works in the bear** (2022 +1.3%; loses every bull year) → shorts are **regime-conditional**, switch on only in confirmed downtrends. (Vindicates not building shorts earlier.) - Caveats: small n/year (18-60), overlapping windows, historical RRAI is a WSB-heavy sampled backfill.

Session findings — sector / calendar / latency (2026-06-01)

- **Sector segmentation rescues the pump signal** (`backtest/sectors.py`). The broad pump was a wash because it *mixed opposites*: **small-cap concentrated pumps FADE** (−14.6%/10d excess, win 22%, **t=−3.93** — pump-and-dumps reliably reverse → a short/avoid), while **biotech pumps are a fat-tailed lottery** (+66% mean, win 48%, t=2.23 — a few catalyst moonshots). Thematic universes (ai/quantum/space/nuclear/meme/frontier) had too few concentrated-pump events to judge.
- **Pump detection is TOO SLOW** (`feature_daily`). Median `novelty_days` at flag = **152**; only **1%** of pumps flagged within 3 days. Root cause: the `min_total≥20` universe filter + trailing z-windows mean a ticker must accumulate ~20 mentions before it's even eligible → we catch *established-ticker spikes*, not fresh pump-and-dumps. Only **33 fresh day-0 bursts** exist in the data. **Fix:** a fresh-burst fast-path that scans raw mentions (ticker's first day with a burst), bypassing the 20-mention gate — for LIVE catching, not just post-hoc.
- **Calendar effects (10y, SPY/QQQ/Gold; absent in AUDJPY-FX):** turn-of-month is real (SPY +0.49 vs +0.20%/5d, QQQ +0.58 vs +0.35, Gold +0.38 vs +0.25); September weak (−0.2%); OpEx week underperforms. Generic equity-flow seasonality → useful as *overlays/filters*, not Reddit alpha.
- **AU sentiment track** building (`au_aggregate_daily` , post-level) to test AU-sentiment → AUDJPY.

★ MASTER BACKTEST vs BUY-AND-HOLD (`backtest/run_all.py` , full 2020-2026, 10d, net of costs)

Every signal judged on EXCESS over its instrument's unconditional drift (buy-and-hold-anytime). | Strategy | n | net% | B&H% | excess | win | verdict | |---|---|---|---|---|---|---| | retail_fear **AUDJPY** | 338 | +0.53 | +0.27 | **+0.25** | 69% | beats (modest, most robust) | | retail_fear SPY | 338 | +0.54 | +0.59 | −0.05 | 61% | **no edge** | | retail_fear Gold | 338 | +0.83 | +0.72 | +0.10 | 57% | marginal | | euphoria_short SPY | 351 | −1.26 | +0.59 | −1.84 | 27% | no edge | | smallcap_pump_fade | 87 | +9.07 | +0.59 | +8.48 | 64% | beats but single-regime + brutal costs | | biotech_catalyst | 91 | +22.2 | +0.59 | +21.6 | 36% | fat-tailed LOTTERY (low win) | | attention_fade | 252 | −0.73 | +0.59 | −1.31 | 40% | no edge |

Recalibration: with a proper buy-and-hold benchmark over the full period, the AUDJPY edge is a **modest +0.25pp/10d** (earlier raw "positive every year" overstated it - that was largely drift). SPY capitulation does NOT beat buy-and-hold. The big-excess equity signals are single-regime (smallcap) or low-win lotteries (biotech). Net: one modest-but-real macro edge, several promising- but-unvalidated screens. Do NOT size up on these as-is.

DATA WE NEED (to validate/strengthen — the binding constraints)

Data	Unblocks	Why we lack it
Intraday prices (we have EOD only)	pump entry/exit timing; fresh-pump fade execution (pumps move intraday)	not collected
Historical penny-stock prices + per-ticker features	multi-regime validation of smallcap_pump_fade & biotech (only 2025-26 exists)	delisted pennies → Yahoo gaps; per-ticker feature_daily only recent
Short interest / float / borrow fee	squeeze setups; realistic short cost/feasibility	FINRA daily short-*volume* in (`scraper/finra.py`); still need SI%/float/borrow (Ortex / FINRA bi-monthly SI)
Options chains / IV	express fades via PUTS (only sane way to short pennies); options-flow signal	not collected
More years of RRAI history	tighten the macro edge (effective n ≈ a few dozen fear episodes)	Reddit/arctic-shift depth ~2017+
Broader author coverage (~20% profiled)	a *PIT-tradeable* deletion/young-account short (thesis already validated forensically; `young_frac` proxy starved)	rate-limited Reddit /about sweep
Independent sentiment sources (StockTwits, news, Discord, crypto + AU subs)	independent episodes → real effective-sample growth without waiting calendar time	not integrated (AU track building)
Real transaction-cost data (per-instrument spreads/borrow)	trustworthy net returns (small-cap costs are modelled/optimistic)	not sourced

Update — complete-data confirmation (2026-06-01)

- US backfill finished: continuous 2020-2026 (2,344 days, 341 capitulation days). **Capitulation-long multi-regime result HOLDS on the complete series** — AUDJPY positive every year incl. 2022 bear (win 60-83%).
- `smallcap_pump_fade` is strong (+12.4% excess short, win 77%) but **only 2025-26 data exists** (n=87) — no historical penny-stock prices/features, so it **can't be multi-regime validated** → stays ADVISORY, NOT promoted to tradeable. (Single-regime; can't be fixed by backfill — penny price history doesn't exist.)

Crypto — coincident, NOT a leading indicator (tested 2026-06-03)

Added CryptoCurrency/Bitcoin/CryptoMoonShots/SatoshiStreetBets (~1yr; Bitcoin 61k posts) and tested whether crypto leads equities / risk-appetite. **It doesn't — it's coincident:** - **Price lead-lag (BTC↔SPY, 2,577d):** same-day corr **+0.17 (t=8.8)** but every lead/lag k≠0 is noise (|t|<1.3) — neither leads. - **Weekend test** (crypto's best shot — BTC trades Sat/Sun while equities are closed; 490 weekends): corr +0.05, **t=1.2** — no lead. - **Crypto sentiment → equity RRAI:** corr **peaks at k=0 (+0.20, t=3.9)** then decays (a lead would peak at k>0). Barely predicts even BTC's *own* next-day return (+0.05). - **Combined crypto+equity RRAI capitulation (1yr overlap, n=55):** does **NOT** help — for the validated AUDJPY edge it *dilutes* it (equity-only excess **+0.55%** > combined +0.45% > crypto-only −0.03%). Crypto sentiment is redundant with (and noisier than) the equity RRAI. - One weak whiff: crypto *euphoria* → mildly negative SPY 3d (t=−2.4, but overlapping windows → true t≈−1.4) — *contrarian*, not leading. Watch, not an edge.

Takeaway: crypto's value is NOT macro timing (it's a redundant coincident risk-appetite read). It's worth keeping only as its own **pump/meme universe** (CryptoMoonShots/SatoshiStreetBets) for the `distribution_short` /pump signals. Do not fold it into the RRAI.

★ Cross-asset generalization map (2026-06-03) — it's a RISK-APPETITE / FEAR-REVERSAL factor

Capitulation-long (rrai≤0.15 & VIX≥18, 10d, excess vs buy-and-hold) tested across asset classes. *Effective window **2018-02 → 2026-04** — the VIX≥18 gate has no signals in the low-vol 2016-17.* The signal fires wherever the instrument is a clean **risk-on** or **fear-bid** play, and is flat everywhere else — the pattern of winners *and* losers is mechanistically coherent (= a real factor):

asset class	works?	evidence
Risk-on FX (AUD/NZD)	✅ **strongest**	AUDJPY t=2.93, NZDJPY t=3.18, AUDCHF 2.83, AUDUSD 2.72, NZDCHF 2.58, ZARJPY 2.32 (EM, noisier)
Safe-haven metals	✅ gold	Gold +0.43% t=2.03; Silver +0.76% t=1.72 (lumpy)
Equity indices (US/AUD/JP)	❌ weak	Nikkei +0.62 t=1.93 (best); S&P t=0.90, Nasdaq 1.13; **ASX 200 t=0.08 (flat)**
Growth commodities	❌ none	Copper t=0.68, Platinum 0.87, **Oil −0.10**
Controls (should be flat)	✅ flat	**AUDNZD (risk/risk) t=1.11**; USDJPY 1.02, CHFJPY 1.37, EURUSD 1.44

Punchlines: (1) the AUD *currency* carries the edge (t=2.9) but the AUD *equity index* (ASX) is flat (t=0.08) — it's an FX/metals phenomenon, not equities. (2) **AUDNZD flat** is the cleanest proof: two risk-on currencies cancel the factor. (3) It's ONE factor, so more instruments = diversification *within* it (diminishing), not new alpha; EM/growth-commodity legs add cost/tail-risk for little gain. **MT5 absolute check** (long-only, ungated, BaseLots=1.0): AUDJPY net +$10.4k (PF 1.14) but AUDUSD ~flat in dollars (it *declined*, so +excess ≠ +absolute) and 60%+ DD → deployable version needs the VIX gate + vol-target sizing.

⚠️ MT5 EXECUTION REALITY CHECK (2026-06-03) — the excess edge does NOT survive naive execution

After fixing the EA's position sizing (ATR fixed-fractional, risk 1%/trade) and hold (trading days), the MT5 tester is finally trustworthy (FX-leg drawdowns fell from ~90% to ~17-20%). It tells a sobering story:

leg (VIX-gated entry, ATR-sized, 10 trading-day hold, **2018-26**)	trades	win%	PF	maxDD%
AUDJPY	67	41.8	**0.85**	20
AUDUSD	67	55.2	**0.91**	17
Gold	67	56.7	**1.01**	93 ⚠️ (no stop → trending losses run)

The validated EXCESS edge (AUDJPY t=2.93, etc.) does NOT translate into a profitable naive long-on-trigger MT5 strategy. Why the gap: 1. **Entry timing.** Python averages over *all* capitulation days (overlapping); the EA enters on the *first* trigger of a cluster — i.e. while the market is *still falling* (the cluster-start = falling knife). The bounce the excess metric sees is *averaged over the cluster*, not capturable by entering at its start. 2. **Excess ≠ absolute.** The FX legs have ~0 unconditional drift, so a positive *excess* (+0.48% vs B&H) nets ~0 in *dollars*. The Python t-stat measures "higher than drift," not "profitable." 3. **No stop.** Holding 10 days with no stop lets losses run (gold DD 93%).

Honest status (before the fix): the edge is a real *statistical* property but was **not deployable** as a naive long-on-trigger trade. Predicted fix: catch the *turn*, not the first fall, + a stop.

✅ RESOLVED — turn-based entry + stop makes it tradeable (2026-06-03)

Added to the EA: **arm on a capitulation signal, then enter on the first up-bar (the turn) within `ArmWindow`, with a `StopATR×ATR` stop**; plus selectable sizing (Fixed / ATR / **Kelly**). MT5 results (VIX-gated → **2018-26**):

symbol	sizing	entry	win%	PF	DD%	Sharpe
AUDJPY	ATR	naive	42.5	**0.77**	9.6	−0.44
AUDJPY	ATR	**turn**	51.4	**1.06**	7.0	0.10
Gold	ATR	naive	53.5	1.25	39.7	0.29
Gold	Fixed	**turn**	55.1	**1.54**	33.3	**0.41**
Gold	ATR	turn	55.1	1.39	44.9	0.40

- **The turn entry works** — it flips AUDJPY from PF 0.77 (losing, naive) to **PF 1.06** (win 42.5→51.4%), and lifts Gold to **PF 1.54 / Sharpe 0.41**. Waiting for the bounce instead of catching the knife is the difference between losing and winning.
- **Gold is the strongest deployable leg** (PF 1.4-1.5, Sharpe 0.4). AUDJPY is now marginally positive (PF ~1.06) — thin but no longer a loser.
- **Sizing verdict: ATR/Fixed win; Kelly OVERSIZES.** Even half-Kelly hit its cap on these thin edges and blew drawdowns to 30% (AUDJPY) / 80% (Gold) with PF≈1.0 — no return benefit. The textbook result: **Kelly is too aggressive for thin/uncertain edges; fixed-fractional ATR (~1%/trade) is the right default.**
- **Still preliminary:** PFs are modest, gold DD is high (33-45%), and this is in-sample on the same series the signal was built on. Forward OOS + a small live demo allocation are the next validation steps. But the basket is now **tradeable, not just a statistical curiosity.**

MT5 parameter optimization (2026-06-03) — in-sample optimize → OOS validate

Ran the **MT5 genetic optimizer** (headless; every pass logged via frames to `rrai_opt_log.csv`; `mql5/RedditMacro_Optimize.ini`) on Gold, sweeping ArmWindow / StopATR / MaxHoldDays, **in-sample 2018-2022**, maximising net profit — then **validated the winners out-of-sample (2023-2026)**:

param set	IS PF / DD	OOS PF / DD / win
profit-max (stop 1.0, hold 5)	1.52 / 55%	1.52 / 42% / 52%
robust (stop 3.0, hold 11)	1.53 / 33%	**1.96 / 18% / 62%**
default (stop 2.0, hold 10)	—	1.75 / 26% / 56%

- **Encouraging:** the signal **didn't badly overfit** — *every* parameter set stayed profitable OOS (PF 1.5-2.0). The gold leg is robust to parameter choice.
- **But "max profit" is the wrong objective:** it picked a tight-stop / quick-exit corner with **2-3× the drawdown** (42-55%) for the *same* PF. **The robust set (wide stop 3×ATR, hold ~11d) wins OOS: PF 1.96, DD 18%, win 62%** — optimise for *risk-adjusted* return (Sharpe/PF-with-DD), not raw profit. *(NB: this 1.96 used ArmWindow=7; the shipped default arm=5 gives ≈1.27 on the same 16 OOS trades — the figure is arm-sensitive on a tiny sample, see the "MT5 deployable-config report" correction.)*
- **ArmWindow ≥3 is irrelevant** (the turn arrives within ~3 bars). The live params worth using on gold: **StopATR≈3, MaxHoldDays≈11**.

Per-instrument optimization (AUDJPY + AUDUSD, same IS→OOS protocol) — the deployable reality

Optimised AUDJPY and AUDUSD the same way (genetic IS 2018-22 → OOS validate 2023-26):

leg	best-robust (IS)	OOS PF / net / DD	verdict
Gold	stop 3, hold 11	**1.96 / +$3.5k / 18%**	✅ **robust — the deployable leg**
AUDJPY	stop 2, hold **24**	1.31 / +$171 / 5%	⚠️ marginal — only +ve OOS with a *long* hold; default (hold 10) is **−ve** (PF 0.87)
AUDUSD	stop 2, hold 5	**0.47 / −$603**	✗ **overfit — fails OOS** (IS PF 1.78 → OOS 0.47)

- **Only Gold robustly survives** realistic execution + OOS. **AUDUSD fails OOS** (its positive *excess* t-stat doesn't translate — excess≠tradeable, again — AUDUSD declined 2023-26 so long-only loses in dollars). **AUDJPY is thin/finicky** (needs hold≈24; the shared-basket default hold underperforms it).

- **Implication for the live basket:** it's really **gold-led**. AUDJPY is a marginal secondary; **AUDUSD (and untested NZDJPY) are not OOS-validated** and should be demo-only or dropped. The single-param-set basket EA is also suboptimal here (gold wants hold 11, AUDJPY wants 24) → **per-instrument params** would be the right enhancement.
- EA defaults set to the **gold-robust** set (StopATR 3 / MaxHoldDays 11 / ArmWindow 5).

▶ Forward-OOS / demo plan (the real test — clock starts now)

In-sample backtests (even OOS-split) can't fully clear overfitting; the honest validation is **forward**: 1. **Deploy on a demo account, gold-led.** Attach `RedditMacro_EA` to **XAUUSD** (and AUDJPY) on a demo, `StrategyTag=retail_fear`, ATR sizing 1%/trade, gold-robust params. Enable WebRequest for the `/signals` URL. 2. **The `/signals` feed already serves the live trigger** (capitulation + VIX + fear_z); the EA mirrors it. From today, every fill is genuine OOS. 3. **Track ~2-3 fear episodes** (a few months) before judging — the edge fires only a handful of times/year. 4. Re-evaluate vs the backtest PF/DD; promote to (small) real size only if forward ≈ backtest.

★★★ RANDOM-ENTRY / BUY-AND-HOLD RECKONING (2026-06-03) — the gold "edge" was drift

The ultimate null test (`backtest/mt5_sim.py::run_benchmarks`): does entering on **capitulation** dates beat entering on the **same number of RANDOM dates** (identical turn/stop/hold/sizing), and does it beat **buy-and-hold**?

leg	strategy net	random-entry (1000×): median / 95th	percentile	buy-and-hold
GOLD	+8.1% (PF 1.52)	**+18.8% / +29.2%**	**4th**	**+240.6%** (DD 21%)
AUDJPY	+6.2% (PF 1.47)	+3.4% / +9.6%	76th	+30.7% (DD 30%)

- **GOLD has NO timing edge — it is WORSE than random** (4th percentile; random beat it 96% of the time) and **buy-and-hold (+240%) crushes it.** The PF 1.52 / OOS 1.27 that survived the excess study, MT5, OOS *and* optimization was **just gold's uptrend**; the capitulation timing actively enters at worse-than-random moments (fear-spike volatility whipsaws the turn-entry + stop). **The "one deployable edge" does not survive this null.**
- **AUDJPY**: 76th percentile — a *weak* positive tilt (better than median random) but **not significant**, and it doesn't beat its own buy-and-hold.
- **Lesson:** the per-trade excess study (which subtracts unconditional drift) *approximated* this null but UNDER-caught it once realistic mechanics (stop/turn/sizing) were added — the mechanics interact with the signal's volatility regime. **Random-entry + buy-and-hold are the non-negotiable final gate.**
- **Net effect on the macro side:** after this test there is **no robustly-deployable macro edge.**

✅ `distribution_short` PASSES the same null (the contrast that matters)

Ran the identical random-entry null on `distribution_short` (`backtest/microstructure.py::random_null`):

test	random median / 95th	dist_short percentile
actual mean short net = **+7.38%** (n=62)	—	—
Null A — random dates, SAME tickers	+0.02% / +3.87%	**100th** (beats all 1000)
Null B — random concentrated pumps	−1.36% / +5.86%	**98th**

- **The opposite of gold.** Shorting the same tickers on *random* dates → ~0%; shorting *random* concentrated pumps → −1.4%. distribution_short's +7.38% sits at the **100th / 98th percentile** → both its **timing** (the price-rolling-over divergence) AND its **selection** (concentrated-pump + euphoric-sentiment gate) add real, non-drift value. **This is the one signal that earns the "real edge" label.**
- **Remaining caveats (so it's not over-trusted):** single-regime (~2025-26, no historical penny prices to multi-regime validate); it **fades in its 2nd half** (date-split t=3.16→0.71); small-cap shorts have squeeze/ borrow/HTB tails → **express via puts**. So: *real, but not yet deployable-with-confidence* — needs forward OOS.

Exhaustive null across EVERY strategy (`backtest/null_all.py`) — and the null's own limitation

Ran the random-entry null on all strategies. **Definitive KILLS (worse than random — the null is decisive here):** `organic_long` (1st pct), `dump_fade_short` (2nd), `deletion_short` (0th), and **every macro capitulation leg** (gold 4th, US500/USTEC 0th, silver 29th, AUDJPY 38th). **Methodological catch:** a naive broad/timing null **over-credits any strategy that enters near a big move** — `pump_long` (100th) "passes" only because it *longs the pump*, yet it regime-flips (t=0.75); `attention_fade` / `fade_organic_short` likewise (tiny +0.3% beat-random- by-selection edges that flip across regimes). So **the null is necessary, not sufficient** — pair it with regime-stability + an *own-universe* control. Only `distribution_short` **passes all of it** (timing null 100th, own-universe null 98th, AND it predicts a *reversal* rather than riding a contemporaneous spike). `smallcap` / `biotech` pass but are single-regime; `AUDUSD` is 96th full-period yet fails OOS. **Bottom line unchanged: one robust edge (`distribution_short`); the macro capitulation complex is confirmed worse-than-random.**

distribution_short — clean-universe validation + forward-OOS log (`backtest/dist_short_oos.py`)

Focusing on the one survivor surfaced a **universe-pollution bug that was halving the edge**. The raw signal set leaked four off-thesis categories — mega-caps (AAPL/NVDA/META/PLTR…), common-word false positives (LIVE/GOOD/TIME/SOAR/BOOM/PR/SI — the *word*, validated as a symbol), ETFs (IAU/IWM/QQQ), and crypto (ETH). Excluding them **a priori, by what the names are (not their returns)**:

universe	n	net	win	t
ALL (polluted)	52	+9.46%	62%	3.13
CLEAN micro-cap only	**25**	**+21.46%**	**84%**	**4.10**
excluded (mega/word/ETF/crypto)	27	−1.65%	41%	−1.72

The thesis predicts exactly this: **manufactured-pump distribution is a micro-cap phenomenon** — Reddit moves price only where liquidity is thin; on mega-caps/ETFs/crypto the same short *loses* (they mean-revert up). The excluded set is a genuine mix (PLTR +6%, INTC −8%, NVDA +2%) netting ~zero, so this is universe-definition, not cherry-picking. **The "recent decay" earlier flagged was a pollution artifact** — on the clean universe the late half is +18.99% (win 85%, t=2.77) vs early +24.13% (t=3.06), i.e. *stable*, not fading. This **supersedes** the date-split "fades to +2.5%, t=0.71" line in the optimization table below — that split was run on the polluted set.

Forward-OOS log is now live (spec frozen 2026-06-03). `dist_short_oos.py` records every fresh clean-universe signal and its realized 10d outcome as the scheduler scrapes; signals dated ≥ freeze are genuine OOS (the spec couldn't be fit to them). 25 in-sample signals seeded (+21.46%, t=4.10); the OOS record starts at zero and builds forward. **This is the one gate distribution_short hasn't faced** — the whole sample still lives in a single ~12-month scraped window, so only forward accumulation can confirm it. Remaining caveats: fat left tail (worst trade JAGU −42% short squeeze → **express via puts**, not naked short); blocklist needs maintenance as new mega-caps / common-word tickers appear.

★★ Comprehensive optimization across ALL strategies — the overfitting reckoning (2026-06-03)

Applied the same protocol (comprehensive genetic IS-optimize 2018-22 → OOS-validate 2023-26, edge params swept, RiskPct fixed at 1%) to **every MT5-tradeable instrument**, and a date-split pseudo-OOS to the Python equity legs. **The result is unambiguous: optimisation overfits almost everywhere; only one edge survives.**

strategy / instrument	IS (optimised)	OOS	verdict
capitulation **GOLD** (robust config, *not* max-optimised)	PF 1.46	**PF 1.27-1.46, DD 22-33%**	✅ **the one deployable edge**
capitulation AUDJPY (optimised)	PF 2.01	**PF 0.99**	✗ overfit — collapsed
capitulation AUDUSD (optimised)	PF 2.47	**PF 0.73**	✗ overfit — lost
capitulation Silver (optimised)	PF 2.85	**PF 0.26 (DD 115%)**	✗ overfit — blew up
capitulation US500 (optimised)	PF 1.91	PF 1.37	~ held but weak signal (excess t≈0.9)
capitulation USTEC	—	—	no usable result
`distribution_short` (equity, date-split)	+11.4% (t=3.16, 1st half)	~~+2.5% (t=0.71, 2nd half)~~	↑ **SUPERSEDED** — fade was pollution; clean universe late-half +19.0% (t=2.77), see above
`smallcap_pump_fade`, `biotech`	strong	**untestable**	single-regime (no penny price history)

- **The tell was visible before OOS:** each instrument's in-sample "optimum" was a *different random-looking corner* (turn 0/1, stop 1-4, hold 7-17, Fixed/Kelly) — no consistent pattern = curve-fit. OOS confirmed it: IS PF 2-2.85 → **OOS PF 0.26-1.37**.
- **Only GOLD survives**, and crucially only with a config chosen for **robustness (a plateau: turn-entry, stop 3, hold 11)**, NOT the max-IS optimum (which would also overfit). distribution_short is the only other thing with real signal, but it fades in its second half and can't be cleanly OOS-tested (single-regime).
- **Net of optimising everything:** the system has **one robustly-deployable edge (gold capitulation) + one promising-but-unvalidated equity short (distribution_short)**. Everything else is overfit, single-regime, or dead. Optimisation did not create edges — it exposed which were real.

Comprehensive optimization (all 7 params, gold IS 2018-22) — profit-max = a leverage trap

Swept **every return-impacting param** (TurnEntry, ArmWindow, StopATR, MaxHoldDays, AtrPeriod, SizingMethod, RiskPctPerTrade), genetic, max-profit. Result:

config	PF	DD%	net
profit-max (turn OFF, stop 1.0, hold 5, **risk 2.0**) — IS	1.54	**67**	+$425k
same — **OOS** 23-26	1.45	**67**	+$15.6k
robust (turn ON, stop 3, hold 11, risk 1.0) — OOS	1.27	**22**	+$1.2k

- `RiskPctPerTrade` **maxed out at the sweep ceiling (2.0)** — it's pure **leverage** (scales profit *and* DD ~linearly), so a profit-max just compounds it to a **67% drawdown**. Risk% is a *risk-tolerance dial, not an edge param* — exclude it from profit optimization.
- The profit-max also flipped to `turn=OFF` **+ tight stop + quick exit** — a different "cut-losses-fast" mechanism that **held OOS (PF 1.45)** but at 67% DD (undeployable). `AtrPeriod` showed no strong effect.
- **Optimising for profit picks a high-leverage / high-DD corner.** The deployable answer is the **risk-adjusted** one: the robust config (turn entry, stop 3, hold 11, risk 1%) — ~same PF at **a third of the drawdown**. EA defaults left on robust.

⚠️ MT5 deployable-config report + honesty correction (2026-06-03)

Ran the current per-instrument config in the MT5 tester:

config	trades	win%	PF	DD%
Gold full 2018-26 (stop3/hold11/1.0%)	63	63.5	1.46	33
Gold OOS 2023-26	16	56.3	**1.27**	22
AUDJPY full 2018-26 (stop2/hold24/0.5%)	54	46.3	1.63	3.1
AUDJPY OOS 2023-26	14	50.0	1.25	2.3

Correction: earlier I quoted **gold OOS PF 1.96**; that used **ArmWindow=7**, whereas the shipped default is **ArmWindow=5**, which gives **PF 1.27** on the same OOS. My "ArmWindow ≥3 is irrelevant" claim held *in-sample* (turns land within ~3 bars in 2018-22) but **NOT out-of-sample** (2023-26 turns sometimes take 4-7 bars). On just **16 OOS trades** this swings PF 1.27↔1.96 — so the honest statement is **gold is OOS-profitable, PF ≈1.3-2.0, arm-sensitive on a tiny sample** (not a hard 1.96). Both legs are +ve IS *and* OOS; AUDJPY adds steady low-DD (2-3%) return as a half-size satellite. Forward demo remains the real test.

✅ Full backtest-suite refresh (2026-06-03, on the extended 2016-26 data, 100% pump-author coverage)

Re-ran every Python backtest module. **Conclusions held — the system is stable.**

module / signal	result on current data	verdict
`run_all` retail_fear **AUDJPY**	excess **+0.21pp**, win 62%	✅ SURVIVOR (core macro edge)
`run_all` retail_fear SPY / QQQ	excess +0.12 / +0.24pp, win 66%	✅ marginal survivors (SPY flipped −0.05→+0.12 on the longer data)
`microstructure` **distribution_short**	**+7.38%/10d, t=2.83**, +ve all 3 regimes	✅ holds (strengthened)
`sectors` smallcap_pump_fade	+8.54% excess short (t=−3.89)	🟡 single-regime
`sectors` biotech_catalyst	+56% but win 42% (t=2.13)	🟡 lottery
`macro_research` euphoria/momentum/fade	all negative-excess except a marginal QQQ down-fade	❌ no edge
`run_all` attention_fade / organic_long	−1.21 / −3.74	❌ dead
`calendar_events` turn-of-month	+1.20%/5d, **t=6.46** (Nov, seasonality)	🟣 real but generic equity overlay, not Reddit alpha
`trend` AUDJPY trend-gate	ungated robust every year; gating thins it	confirms AUDJPY stays **ungated**
`trend` euphoria_short	only +ve in the 2022 bear	confirms **parked**

Caveat (the gap that matters): these are the *statistical* (excess-vs-B&H) edges; the macro excesses are thin (+0.1-0.2pp). The *tradeable* reality from the MT5 + OOS work is narrower — only **gold** survived realistic single-entry execution out-of-sample (PF 1.96/DD 18%), AUDJPY marginal, AUDUSD/NZDJPY failed. So: statistically real = capitulation-long + distribution_short; realistically deployable = **gold-led**, forward-demo pending.

Changelog

- **2026-06-03** — ★ **Multi-regime validation + frothy-retail regime added to the scoreboard.** Deep-backfilled the corpus to 2021 (`archive_backfill_deep` , 5 micro-cap subs, 330k+ posts; DuckDB research store, 9.6× faster). Three-regime test (`backtest/regime_test.py` , `backtest/distribution_short.py`): distribution_short is **regime-CONDITIONAL** — 2021 mania null 46th (DEAD, re-squeezes), 2022 bear +18.7%, 2025-26 +20.96% (null 100th). **MT5-validated on broker OHLC** (`scripts/mt5_dist_short_backtest.py`): 36/37 names tradeable, +11.02%, pattern replicates. **Cap-tier:** works micro→small→mid, reverses on large/mega; strongest in liquid $5-15 band (t=5.26). Added `frothy_retail_regime` as a tracked regime-filter idea (detector unbuilt — froth-index failed).

- **2026-06-03 — ★ distribution_short clean-universe + forward-OOS log (`backtest/dist_short_oos.py`).** Found the signal universe was polluted with mega-caps / common-word false-positive tickers / ETFs / crypto — excluding them a priori **doubled the edge to +21.46% (win 84%, t=4.10)** and revealed the earlier "recent decay" was a pollution artifact (clean late-half +19.0%, t=2.77 — stable). Froze the clean spec and started the forward-OOS log: every fresh signal ≥ 2026-06-03 is genuine out-of-sample, accumulating as the scheduler scrapes. The last untested gate is now ticking.
- **2026-06-03 — Exhaustive random-entry null on ALL strategies (`backtest/null_all.py`).** Definitive kills (worse than random): organic_long, dump_fade_short, deletion_short, and every macro capitulation leg (gold 4th, US500/USTEC 0th). Key methodological finding: the naive null **over-credits momentum/selection strategies** (pump_long 100th but regime-flips; attention_fade/fade_organic 99th but flip) — it's necessary, not sufficient; must pair with regime-stability + own-universe control. Only **distribution_short passes everything**. Bottom line unchanged: one robust edge, macro capitulation confirmed worse-than-random.
- **2026-06-03 — distribution_short PASSES the random-entry null — the one real edge.** Same null applied (`microstructure.random_null`): vs random dates on the same tickers → **100th percentile**; vs random concentrated pumps → **98th**. So its timing (divergence) AND selection (gate) add real, non-drift value — the exact opposite of gold (4th pct). Still single-regime + fades in 2nd half + small-cap → puts, so real but not yet deployable-with-confidence. The system's honest state: **ONE real edge (distribution_short), no deployable macro edge.**
- **2026-06-03 — ★ RANDOM-ENTRY RECKONING — the gold edge was drift.** Added random-entry Monte Carlo (1000×, same #entries + mechanics, random dates) + buy-and-hold to `mt5_sim`. **GOLD capitulation is at the 4th percentile of random** (worse than random; B&H +240% vs strategy +8%) → the PF that survived excess/MT5/ OOS/optimization was just **gold's uptrend**, with the signal timing it *worse than random*. AUDJPY only 76th (weak, n.s.). **No deployable macro edge survives this null** — `retail_fear` downgraded to ❌. distribution_short (has a built-in control) is the only signal with a null-beating result, but is single-regime. Random-entry + B&H are now the mandatory final gate.
- **2026-06-03 — Full backtest suite + optimization now runs LOCALLY in Python.** Added a local full-grid optimizer to `mt5_sim` (IS→OOS, on the MT5 broker OHLC, ~7k sims in seconds) + series caching. Re-ran everything locally: the MT5-replica reproduces the reckoning (only GOLD holds OOS, max-IS PF 2.19; AUDJPY/Silver/US500 overfit→fail), and the Python suite is unchanged (capitulation_long SURVIVOR SPY/QQQ/AUDJPY, distribution_short +7.4% t=2.83, smallcap/biotech single-regime, turn-of-month overlay). No MT5 dependency for backtesting any more — MT5 is just the live execution shell.
- **2026-06-03 — Pulled MT5 broker OHLC into Python → the replica now MATCHES the MT5 tester.** Used the `MetaTrader5` Python package (`scripts/export_mt5_ohlc.py`) to export D1 OHLC straight from the terminal (the exact tester data) for all 6 instruments → `mt5_ohlc` table. `mt5_sim.py` now uses true-range ATR + intraday stops and **reproduces MT5** (gold OOS PF **1.27 == 1.27**; AUDUSD/silver fail OOS in both). The strategy now runs fully in Python on the real broker data, MT5-free. Residual gaps: closed-trade vs floating-equity DD, and no spread/swap (FX) — both addressable.
- **2026-06-03 — Python replica of the MT5 EA (`backtest/mt5_sim.py`) — cross-check.** Re-ran the capitulation legs in Python (turn entry / ATR stop / risk sizing). Qualitative verdicts reproduce (gold +ve, **silver overfits 2.02→0.76 OOS**, all legs thin PF ~1.0-1.5) — BUT the prices table is **close-only**, so DD is badly understated (Python 4-7% vs MT5 22-67%) and stops are approximate. Confirms the edge story; NOT reliable for drawdown. **Data gap to close: OHLC prices** would make the Python backtest MT5-accurate.
- **2026-06-03 — Optimized ALL strategies (IS→OOS) → the overfitting reckoning.** Comprehensive genetic optimization of capitulation-long on every MT5 instrument (AUDJPY/AUDUSD/Silver/US500/USTEC) + date-split for distribution_short. **Every per-instrument optimum overfit and failed OOS** (IS PF 2.0-2.85 → OOS 0.26-1.37); each "best" was a different curve-fit corner. **Only GOLD survives** (robust config, OOS PF 1.27-1.46). distribution_short real but fades (1st-half t=3.16 → 2nd-half t=0.71). smallcap/biotech untestable (single-regime). Conclusion: one deployable edge (gold) + one unvalidated (distribution_short).
- **2026-06-03 — Comprehensive optimization (all 7 params) → profit-max is a leverage trap.** Swept TurnEntry/ArmWindow/StopATR/MaxHoldDays/AtrPeriod/SizingMethod/RiskPct (genetic, gold IS). RiskPct maxed to the 2.0 ceiling (it's leverage, not edge) → 67% DD; profit-max flipped to turn-OFF/tight-stop/quick-exit (+$425k IS, held OOS PF 1.45 but 67% DD = undeployable). The risk-adjusted answer stays the robust config (turn/stop3/hold11/risk1%): ~same PF at 22-33% DD. Lesson: optimise risk-adjusted, set RiskPct by tolerance.
- **2026-06-03 — MT5 deployable-config report + honesty correction.** Ran the current per-instrument config: Gold full PF 1.46 / OOS 1.27 (DD 22-33%); AUDJPY full PF 1.63 / OOS 1.25 (DD 2-3%). Both legs +ve IS *and* OOS. **Corrected an overstatement:** the earlier "gold OOS PF 1.96" used ArmWindow=7; the shipped default arm=5 gives PF 1.27 on the same 16 OOS trades. "ArmWindow≥3 irrelevant" held IS but NOT OOS — so gold OOS is PF ≈1.3-2.0, arm-sensitive on a tiny sample (not a hard 1.96).
- **2026-06-03 — Full backtest-suite refresh.** Re-ran run_all / fear_gate / microstructure / sectors / calendar_events / trend / macro_research on the extended 2016-26 data. All conclusions held: capitulation-long a consistent SURVIVOR (AUDJPY +0.21 / SPY +0.12 / QQQ +0.24 excess), distribution_short +7.38% t=2.83, smallcap/biotech single-regime, euphoria/momentum/attention dead, turn-of-month a generic overlay. System stable.

- **2026-06-03 — Gold-led basket + per-instrument EA params.** Trimmed live `INSTRUMENTS` to the OOS survivors: **XAUUSD (primary) + AUDJPY (small secondary)**; dropped AUDUSD (OOS PF 0.47) and NZDJPY (no OOS). EA live basket now uses **per-leg** StopATR/MaxHoldDays/Risk (gold 3/11/1.0%, AUDJPY 2/24/0.5%) instead of one shared set — so each instrument runs at its own OOS optimum. Recompiled; tester path unchanged (gold full-period PF 1.46 sanity-checked).
- **2026-06-03 — Optimized AUDJPY+AUDUSD too → only GOLD survives OOS; set gold defaults; forward plan.** Same IS→OOS protocol: Gold robust (OOS PF 1.96/DD 18%), AUDJPY marginal (OOS PF 1.31, needs hold≈24, default hold-10 goes −ve), **AUDUSD overfits & fails OOS (1.78→0.47)** — its +excess t-stat doesn't trade (it declined). So the live basket is **gold-led**; AUDUSD/NZDJPY demo-only/drop. Set EA defaults to the gold-robust set (StopATR 3/hold 11/arm 5, recompiled). Added a forward-OOS/demo deployment plan.
- **2026-06-03 — MT5 optimizer run (gold): in-sample optimize → OOS validate.** Genetic sweep of entry/exit params, profit-maximised IS (2018-22), validated OOS (2023-26). Signal didn't badly overfit (all sets OOS-profitable, PF 1.5-2.0) but max-profit = a high-DD corner (42-55%); the **robust set (stop 3×ATR / hold 11) is best OOS: PF 1.96, DD 18%, win 62%**. Added headless frame-logging + `RedditMacro_Optimize.ini`.
- **2026-06-03 — Turn entry + stop makes it tradeable; sizing methods backtested.** Added to the EA: arm on capitulation → enter on the first up-bar (turn) within ArmWindow, with a StopATR×ATR stop; plus selectable sizing (Fixed/ATR/Kelly). The turn entry **rescues the strategy**: AUDJPY PF 0.77→1.06, Gold PF 1.25→1.54 (Sharpe 0.41). Sizing: ATR/Fixed best; **Kelly oversizes** (caps out on thin edges → 30-80% DD, no return gain). Gold is the strongest leg. Still in-sample/preliminary → forward OOS + demo next.
- **2026-06-03 — MT5 reality check: excess edge ≠ tradeable.** Fixed the EA (ATR fixed-fractional sizing
- trading-day holds; compiled clean), which collapsed the fixed-lot drawdown artifact (90%→17-20% on FX) and revealed the truth: under realistic single-entry execution the capitulation longs are breakeven-to- negative (AUDJPY PF 0.85, AUDUSD 0.91, gold 1.01 but 93% DD). The validated *excess* edge doesn't survive naive long-on-trigger execution — gap is entry timing (cluster-start = falling knife) + excess≠absolute + no stop. Needs a turn-based entry + stop to deploy. Basket downgraded to research-validated, not live-tradeable.
- **2026-06-03 — Trimmed live basket to tradable-alpha only.** Per the cross-asset map, kept the four legs with significant excess (t≥2): **AUDJPY (2.93), NZDJPY (3.18), AUDUSD (2.72), Gold (2.03)**. Dropped US500/USTEC (t≈1) and Silver (1.72, lumpy); parked `euphoria_short` (traded the dropped indices, −1.84 excess). `signals.py` INSTRUMENTS + MT5 EA basket trimmed 7→4; GATE_TREND cleared. Deployed live.
- **2026-06-03 — Cross-asset generalization map + MT5 FX backtest.** Capitulation-long tested across FX/indices/commodities: it's a **risk-appetite/fear-reversal factor** — strong in risk-on FX (AUD/NZD, EM-ish), works in gold/silver, **flat in equity indices (incl. ASX t=0.08) and growth commodities (copper/oil)**; AUDNZD control flat. MT5 tester (D1, 2016-26, ungated): AUDJPY +$10.4k/PF 1.14, AUDUSD ~flat-$ (declined, +excess only), 60% DD → needs VIX gate + vol-target to deploy.
- **2026-06-03 — Cross-sectional FX validation → broadened the basket.** Tested capitulation-long across the yen-cross + risk-pair complex (2018-26, VIX-gated). The edge is a **risk-on-currency factor**: every AUD/NZD pair is significant (NZDJPY t=3.18, AUDJPY 2.93, AUDCHF 2.83, AUDUSD 2.72) *independent of the funding leg*, while the no-risk-leg controls are flat (USDJPY t=1.02, CHFJPY 1.37, EURUSD 1.44) — the controls failing is the proof it's real, not data-mining. It's NOT a JPY thing (JPY is just a clean funding leg; Nikkei works as high-beta risk, not as "Japanese"). **Broadened the live `retail_fear` FX basket to AUDJPY + NZDJPY + AUDUSD** (`signals.py` + MT5 EA).
- **2026-06-03 — Deletion short DIED at full coverage; AUDJPY re-confirmed; sweep de-crypto'd.** (1) Targeted-profiled all 487 equity pump-authors (25%→100% relevant coverage). The deletion short **flipped negative** — gone≥0.20 → −19.3%/10d (n=87), young_frac short −39%→−87% — dominated by squeeze tails (IXHL −625%) the 25% sample had missed. The +14.3% was a **coverage-bias artifact**. Dead as a stock short; puts-only. `distribution_short` (price-rolling-over filter) is the survivor. (2) **AUDJPY capitulation re-validated**: VIX-gated +0.48%/10d **t(exc)=2.93 (2018-26**, the gate excludes calm 2016-17), **+ve 8/9 years**; ungated +0.26% t=2.52 (2016-26) (2018 the lone falling-knife miss). Now the most significant edge in the system; unaffected by author work (pure macro). (3) `get_authors_needing_age` now skips crypto-only authors + front-loads ticker-mentioners.
- **2026-06-03 — Crypto leading-indicator test: COINCIDENT, not leading.** Price lead-lag BTC↔SPY ≈0 (same-day +0.17 but k≠0 noise), weekend test t=1.2, crypto sentiment coincident with equity RRAI (peaks k=0). Combined crypto+equity RRAI does NOT improve capitulation — *dilutes* the AUDJPY edge (+0.55%→+0.45%). Crypto's value = its own pump/meme universe, not macro timing. (See "Crypto — coincident" section.)
- **2026-06-03 — RRAI extended to 2016-06 (525 caps, +190) → fear-gate STRENGTHENED.** Backfill landed (aggregate_daily 2016-06→2026, 3,653 days; +2018 Volmageddon/Q4 episodes). Re-ran `backtest.fear_gate`: fear-only excess grew SPY +0.32→+0.57, QQQ +0.41→+0.80 (effect *grew* OOS = real-edge signature); episode-level now **significant on BOTH legs and they agree** (SPY t=2.77, QQQ t=2.37, k=63) — the prior SPY≠QQQ noise-floor flag is resolved. Day-level still marginal (1.6-1.9). The LIVE `trend+fear` gate is still ≈0 (SPY +0.10/QQQ +0.11) ⇒ **drop the trend filter, use fear-only**. AUDJPY confirmed best ungated.

- **2026-06-02 — Power analysis + RRAI history extension (in progress).** Quantified the fear-gate's data need: the 94 fear-days are ~40 independent episodes; day-level it's t≈0.6 (needs ~9× more obs for $p<0.05$, ~19× for power — impractical), but episode-level it's t=2.28 (SPY)/1.85 (QQQ) — significant-but- fragile (COVID-cluster-driven; SPY≠QQQ ⇒ noise floor). Cheapest lever = more *independent fear episodes* via **backward** history (Wikipedia fear data already reaches 2015). Extended study prices to max and **kicked off the RRAI aggregate backfill 2016-06→2020** (wallstreetbets/stocks/pennystocks) to add the 2018 Volmageddon + Q4-2018 episodes; will re-run `backtest.fear_gate` on the longer series when it lands.
- **2026-06-02 — Committed EXCESS study settles the fear-gate (`backtest/fear_gate.py`).** The proper frame (mean excess vs buy-&-hold, the +0.36pp's own metric). Three findings: (1) the +0.36pp **reproduces** as a point estimate (SPY +0.32 / QQQ +0.41 fear-only excess) — not fabricated; (2) it's **not significant** (excess t≈0.6, n≈94); (3) the LIVE **trend+fear** combo is *worse* — dilutes SPY (→+0.09), flips QQQ negative (→−0.20) — because the trend filter strips the downtrend bounces that carry the excess. Reconciles the MT5 "loss": MT5 sums absolute P&L (COVID-cluster dominates), excess means-averages — so the earlier "drag" was a frame artifact. Gate stays 🟡; fear-only > trend+fear for excess; AUDJPY stays ungated (the only significant edge). Exporter refactored to share the study's `capitulation_sets()` (MT5 + Python test identical dates).
- **2026-06-02 — MT5 backtest of the fear-gate → it's a DRAG, not a rescue.** Extended `--export-signals` to emit 4 point-in-time capitulation date-sets (baseline / `fear_z≥0.5` / 200-DMA-trend / trend+fear) and ran each through the MT5 Strategy Tester on US500 (2020-26) via `RedditMacro_US500_*.ini`. Adding `fear_z` made absolute returns *worse* (alone −$1045, with trend −$372 vs trend-only +$21) — fear spikes fire long into tops (Feb-2020). Only the trend filter helps (DD 13.6→7.1%). Caveat: MT5 = absolute long-only, the +0.36pp was *excess* — so not a direct refutation, but the gate is downgraded to 🟡 doubtful and needs a committed *excess-based* study to settle (maybe drop `fear_z`, keep trend).
- **2026-06-02 — Doc-completeness pass.** Audited the doc against the live registry + all backtest modules. Found one missing edge — the **Wikipedia fear-confirmation gate** (live: gates the equity legs of `retail_fear`, `fear_z≥0.5`, +0.36pp rescue) — now documented, *honestly flagged as not yet committed-backtested*. Also added the full live-screen list (`fresh_pump_alert`, `coordination_flag`, ...) to the scoreboard and recorded the broad "attention-leads-price" long as a failed up-leg test.
- **2026-06-02 — New PIT-tradeable short found: `distribution_short`** (`backtest/microstructure.py`). Built the missing **new-author recruitment** feature and tested demand-quality timing signals. The decisive lesson: microstructure timing (recruitment-exhaustion / deceleration / divergence) is **worthless on the broad attention universe** (it drifts flat-to-up, costs dominate) but **powerful inside the concentrated/manufactured subset** — exactly where the deletion edge lives. Winner = concentrated pump + euphoric sentiment + price already rolling over → **+7.2%/10d net, t=2.70 (date-clustered t=2.52, 47 indep. dates), dose-response, survives 2× costs**; control loses −4.1%. Unlike the deletion gate it uses **no hindsight** → tradeable. Recorded negatives (broad timing, recruitment-alone, up-leg ride) to avoid re-runs.
- **2026-06-02 — Deletion-gated short forensically validated.** Author coverage 8%→20% (2,973 gone). Re-backtest: `gone_frac≥0.20` → +14.3%/10d short (83% win, t=2.57, +ve all regimes); the *control* (un-gated concentrated pumps) *loses* −10.8% — so the deletion gate flips the trade, confirming the A2 thesis. **Caveat:** `gone_frac` is hindsight (no Reddit deletion timestamp) → not yet PIT-tradeable; the `young_frac` PIT proxy is still too sparse (n=3–7). Also shipped `scraper/finra.py` (FINRA Reg SHO daily short-volume, 475k rows) + z-score `squeeze_screen()` — learned the *absolute* short-ratio is market-maker noise (median 0.49), so the screen uses per-ticker z-score anomaly + chatter.
- **2026-06-01** — Instrument scan (17 MT5 instruments). `retail_fear` generalises across the risk-on basket; commodities express it best (Silver +2.2pp lumpy, Gold +0.6pp robust), AUDJPY steadiest. Oil/crypto dropped. Basket = AUDJPY + Gold + indices + small Silver; EA + feed updated.
- **2026-06-01** — Macro R&D + MT5 framework. Benchmarked candidates vs buy-and-hold: only the `retail_fear` contrarian family beats it (best AUDJPY +0.47pp/10d excess); momentum/froth = no edge, all shorts fail. Built `SignalClient.mqh` + `RedditMacro_EA.mq5` (strategy-selectable), multi-strategy `/signals` feed; retired `RedditRRAI_EA.mq5`.
- **2026-06-01** — Initial document. Phase 0 (feature store + friction-aware backtester) shipped. RRAI capitulation-long validated (best via AUDJPY, VIX-gated); pump-ride/naive-dump-short/organic-long failed; fade-organic marginal; deletion-gated short awaiting author coverage (8%). `/signals` + MT5 EA live.